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May 11, 1976
SUPERSEDING
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February 28, 1972

FEDERAL SPECIFICATION

WETTING AGENT, LIQUID, PHOTOGRAPHIC GRADE

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification establishes the requirements for liquid wetting agents. The wetting agents shall be added to water to prepare a final rinse bath for processed photographic films, plates, and papers to prevent the formation of water marks and streaks.

1.2 Classification. Liquid wetting agents shall be supplied in plastic containers in the following unit sizes, as specified (see 6.2):

Unit sizes

4 oz.
16 oz.
1 gallon

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

FSC 6750

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Military Specifications

MIL-F-26615 - Fixing Bath, Conventional Photographic, Acid Hardening.

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Concentration. For acceptance test purposes, the standard working solution shall consist of 1 part liquid wetting agent and 1600 parts of distilled water. All wetting agents to be tested shall be mixed with distilled water to equal the standard working solution.

3.2 Solubility. When prepared in accordance with 4.4., the wetting agent, shall mix readily with the water, without forming a precipitate, to form a clear working solution. Solution clarity shall not be affected when tested as specified in 4.4.1.

3.3 Surface tension. The surface tension of the working solution shall not be more than 40 dynes per centimeter when tested in accordance with 4.4.2.

3.4 Wetting. When tested as specified in 4.4.3, the entire area of both sides of the photographic material shall be uniformly wet.

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3.5 Residue. When tested in accordance with 4.4.4, there shall be no evidence of watermarks, streaks, or other residue.

3.6 Label durability. When tested as specified in 4.4.5, the printing on the label shall show no evidence of smearing or fading, and the label shall remain firmly attached to the container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lots. A lot for inspection purposes shall consist of all of the wetting agent offered for delivery at one time.

4.3 Sampling. Samples for inspection of the product and fill of the containers shall be selected in accordance with the applicable provision of MIL-STD-105 at inspection level II and an acceptable quality level of 2.5.

4.4 Quality conformance inspection. One gallon of working solution, (see 3.1), shall be prepared by mixing one part wetting agent with 1600 parts of distilled water. Failure to meet the requirements as specified in 3.2 shall be cause for rejection. The working solution shall be used for the quality conformance tests specified herein.

4.4.1 Solubility test. One liter of the working solution shall be stored in a covered clear glass container for one week at temperatures not to exceed 38°C. The clarity of the solution shall then be noted. The chemical contaminants listed below shall then be added to the solution in the following order:

1 gram of calcium chloride
1 gram of aluminum potassium sulfate
10 ml of standard solution #1 as specified in MIL-F-26615

The effect of each chemical on the clarity of the solution shall be cause for rejection.

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4.4.2 Surface tension test. The surface tension of the working solution shall be measured at 20.0° ± .5°C with a DuNouy Tensionometer (or equivalent instrument) in accordance with the instrument manufacturer's directions. Failure to meet the requirements specified in 3.3 shall be cause for rejection.

4.4.3 Wetting test. One photographic glass plate, one 4 by 5-inch sheet of photographic film, and one 8 by 10-inch sheet of sensitized paper shall be developed, fixed, and washed in accordance with the recommendations of the manufacturer. Immediately following the wash cycle, the photographic materials shall be totally immersed in a bath of the wetting agent working solution for approximately 30 seconds. Immediately following the wetting agent bath the photographic materials shall be suspended by one corner allowing the excess working solution to drain. The plate, film, and paper shall be visually inspected to determine that they are uniformly wet. Failure to meet the requirements specified in 3.4 shall be cause for rejection. The photographic materials shall be air dried and used in the residue test (see 4.4.4).

4.4.4 Residue test. A drop of unused wetting agent working solution shall be placed on a clean dry ferrotype plate and allowed to dry. The ferrotype plate and the dried photographic materials obtained from the wetting test (see 4.4.3) shall be examined for evidence of watermarks, streaks, or other residue. Failure to meet the requirements specified in 3.5 shall be cause for rejection.

4.4.5 Label durability test. Each sample container shall be immersed in the working solution of 21.0° ±.5°C for 30 minutes. The label shall be examined to determine whether the printing has faded or smeared, and that the label remains firmly attached to the container. Failure to meet the requirements specified in 3.6 shall be cause for rejection.

4.5 Inspection of preparation for delivery. Packaging and packing shall be tested and examined in accordance with the applicable provisions of section 5 of the respective packaging and packing specifications referenced herein.

5. PREPARATION OF DELIVERY

5.1 Packaging. Each package shall be a close-fitting box conforming to PPP-B-636, type CF, class domestic. Boxes shall be fitted with slotted fiberboard partitions and double walled corrugated fiberboard top and bottom pads. Closure of boxes shall be in accordance with method II of the appendix to PPP-B-636. Unit containers shall be packaged in quantities, as follows:

Unit Size	Units per Package
4 oz.	24
16 oz.	12
1 gallon	4

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5.2 Packing. Packing shall be A, B, or C, as specified (see 6.2).

5.2.1 Level A. Wetting agent, packaged as specified in 5.1, shall be packed in a class 3, style 2 wirebound wood box in accordance with PPP-B-585 (see 6.2).

5.2.2 Level B. Wetting agent, packaged as specified in 5.1 shall be packed in a class 1, style 2 wirebound wood box in accordance with PPP-B-585 (see 6.2).

5.2.3 Level C. Wetting agent shipped directly from the manufacturer to the consumer shall be packed to insure carrier acceptance and safe delivery at the destination (see 6.2).

5.3 Marking. Unit packaging and packing containers shall be marked in accordance with the applicable provisions of MIL-STD-129 for military shipments and Fed. Std. No. 123 for civil agencies. Each unit container shall be permanently labeled with the name of the material, the quantity contained, and the manufacturer's name or trademark. In addition, each unit container shall be legibly and permanently marked with complete directions for use, explaining clearly the proper method of mixing and the amount of wetting agent required to treat one gallon of water.

6. NOTES

6.1 Intended use. The wetting agent covered by this specification is intended for use in preparation of a final rinse bath for photographic films, plates and papers to prevent the formation of water marks and streaks.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Unit size and quantity of wetting agent required (see 1.2 and 5.1).
(c) Level of packing to be specified by the procuring agency (see 5.2).

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